

## Exercise 1: SQL Fundamentals (Select - Where) - 28 April <sup>①</sup>

01. Retrieve all columns from the employees table

```
SELECT *  
FROM employees;
```

| id | first_name            | last_name | department | salary | hire_date  | city         |
|----|-----------------------|-----------|------------|--------|------------|--------------|
| 1  | Alice                 | Green     | IT         | 70000  | 2020-01-10 | Johannesburg |
| 2  | Brian                 | Lee       | HR         | 45000  | 2019-03-22 | Cape Town    |
| 3  | Cathy                 | Zulu      | Finance    | 65000  | 2018-07-18 | Durban       |
| 4  | David                 | MoKoena   | Marketing  | 50000  | 2021-11-05 | Pretoria     |
| 5  | <del>Maidoo</del> Eva | Naidoo    | IT         | 72000  | 2017-09-30 | Johannesburg |

02. Find all unique departments

```
SELECT DISTINCT (department)  
FROM employees;
```

| department |
|------------|
| IT         |
| HR         |
| Finance    |
| Marketing  |

03. Retrieve first and last name ordered by salary descending

```
SELECT first_name,  
       last_name  
FROM employees  
ORDER BY salary DESC;
```

| first_name | last_name |
|------------|-----------|
| Eva        | Naidoo    |
| Alice      | Green     |
| Cathy      | Zulu      |
| David      | MoKoena   |
| Brian      | Lee       |

04. Retrieve the top 3 highest-paid employees

```
SELECT id,
       first_name,
       last_name,
       salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

| id | first_name | last_name | salary |
|----|------------|-----------|--------|
| 5  | Eva        | Naidoo    | 72000  |
| 1  | Alice      | Green     | 70000  |
| 3  | Cathy      | Zulu      | 65000  |

05. Find employees in the IT department

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department = 'IT';
```

| id | first_name | last_name | department |
|----|------------|-----------|------------|
| 1  | Alice      | Green     | IT         |
| 5  | Eva        | Naidoo    | IT         |

06. Find employees in Finance with salary > 60000

```
SELECT id,
       first_name,
       last_name,
       department,
       salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;
```

| id | first_name | last_name | department | salary |
|----|------------|-----------|------------|--------|
| 3  | Cathy      | Zulu      | Finance    | 65000  |

(3)

07. Find employees in HR or Marketing

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department = 'HR' OR department = 'Marketing';
```

| id | first_name | last_name | department |
|----|------------|-----------|------------|
| 2  | Brian      | Lee       | HR         |
| 4  | David      | Mokoena   | Marketing  |

08. Find employees not in IT

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE employees != 'IT';
```

| id | first_name | last_name | department |
|----|------------|-----------|------------|
| 2  | Brian      | Lee       | HR         |
| 3  | Cathy      | Zulu      | Finance    |
| 4  | David      | Mokoena   | Marketing  |

09. Find employees in IT, HR or Finance using IN  
 SELECT id,  
     first\_name,  
     last\_name,  
     department  
 FROM employees  
 WHERE department IN('IT', 'HR', 'Finance');

| id | first_name | last_name | department |
|----|------------|-----------|------------|
| 1  | Alice      | Green     | IT         |
| 2  | Brian      | Lee       | HR         |
| 3  | Cathy      | Zulu      | Finance    |
| 5  | Eva        | Naidoo    | IT         |

10. Find employees in IT with salary > 65000 and city Johannesburg

SELECT id,  
     first\_name,  
     last\_name,  
     department,  
     salary  
 FROM employees  
 WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';

| id | first_name | last_name | department | salary | city         |
|----|------------|-----------|------------|--------|--------------|
| 1  | Alice      | Green     | IT         | 70000  | Johannesburg |
| 5  | Eva        | Naidoo    | IT         | 72000  | Johannesburg |